PROJECT BRIEF

Surrogate-Assisted Multi-Objective Optimization of RC Structures for Cost and Embodied Carbon

A Constrained NSGA-II Framework using OpenSeesPy-Trained ML Surrogates

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1. Executive Summary

This project builds a computational design tool that generates cost-vs-embodied-carbon trade-off options for reinforced concrete (RC) structural elements, instead of a single “optimal” design. Structural response (deflection, member utilization) is simulated in OpenSeesPy across a large sample of design combinations; a machine-learning surrogate model is trained on that data to predict structural performance far faster than repeated finite-element analysis; and a constrained NSGA-II optimizer (via the Python pymoo library) searches the design space against that surrogate to produce a Pareto front of designs — each one a different balance between construction cost and embodied carbon, all satisfying IS-code strength and serviceability limits. A subset of the final Pareto-optimal designs is re-verified with real OpenSeesPy analysis to confirm the surrogate did not mislead the search.

The project sits at the intersection of structural engineering, optimization, and applied machine learning — chosen deliberately because it lets me apply the ML tooling I already work with (Python, scikit-learn/XGBoost, multi-objective optimization) inside my home discipline, rather than treating the major project as disconnected from the rest of my technical work.

2. Motivation & Problem Statement

RC design in practice is still dominated by single-objective, code-minimum thinking — engineers size members to satisfy strength and serviceability at least cost, with embodied carbon treated as an afterthought if at all. Life-cycle carbon accounting is becoming a real constraint on construction (green building codes, client ESG requirements), but there is no simple way for a designer to see, at the design stage, what a lower-carbon alternative would cost. This project produces that view directly: a Pareto front showing, for a given structural element or frame, the full range of cost-carbon trade-offs available — not a single black-box “recommendation.”

A second, practical motivation: naive multi-objective optimization against a full finite-element model is computationally expensive — NSGA-II typically needs thousands of evaluations to converge, and each OpenSeesPy run has non-trivial cost when repeated at that scale. Training a fast ML surrogate on a bounded batch of FE runs, then optimizing against the surrogate, is the standard way this is handled when the true evaluation function is expensive — the same pattern used in recent ML-assisted concrete mix design work.

3. Literature Position — Honest Novelty Statement

This is not being framed as a novel algorithm. NSGA-II-based multi-objective optimization of RC frames for cost has been published since at least 2016, and ML-surrogate-assisted NSGA-II frameworks jointly optimizing cost and carbon already exist in adjacent domains (e.g. concrete mix design). The differentiation of this project is deliberately narrow and stated up front so expectations are correct from day one:

- Application of the surrogate-assisted NSGA-II pattern specifically to RC member/frame sizing (not mix design), with cost AND embodied carbon as co-equal objectives.
- Explicit handling of India-specific constraints: IS 456 / IS 800 / IS 875 design basis, discrete Indian standard section and rebar-diameter sizing, and embodied-carbon factors sourced or adapted for Indian materials rather than assumed from UK/EU databases (ICE) by default.
- A verification step that re-checks surrogate-selected Pareto designs against ground-truth OpenSeesPy analysis — explicitly reported as an accuracy/reliability result, not assumed.

This framing is meant to be defensible in a viva: the contribution is a carefully validated, India-contextualized application and workflow, not a claim of a new optimization method.

4. Objectives

Primary

- Produce a validated Pareto front of RC design options trading off construction cost against embodied carbon, for a defined structural scope (Section 5), all satisfying IS-code strength and serviceability constraints.

Secondary

- Quantify how much computational cost the ML surrogate saves versus running NSGA-II directly against OpenSeesPy (evaluations required, wall-clock time).
- Quantify surrogate prediction error against ground-truth OpenSeesPy results on held-out and Pareto-boundary designs.
- Document which Indian-context embodied carbon data was available vs. had to be estimated/adapted, as a usable reference for future students.

5. Scope & Assumptions — For Guide's Confirmation

The items below are proposed defaults, not fixed decisions — flagged explicitly because they are the guide's call and change the effort required:

- Structural typology: RC moment-resisting frame (proposed: G+3 to G+5, regular plan) rather than a single isolated beam/column, to keep the result relevant to real design decisions.
- Loading: gravity (dead + live per IS 875) as the base case; seismic loading (IS 1893 / IS 13920 detailing) as an optional extension if time permits — recommend deferring seismic to keep Phase 1–4 scope achievable in one project cycle.
- Design variables: column/beam cross-section dimensions, longitudinal reinforcement ratio, concrete grade (M20–M40), steel grade (Fe415/Fe500) — full list in Section 8.
- Design code basis: IS 456:2000 (RC design), IS 800:2007 (if steel members included), IS 875 (loads).
- Out of scope (explicitly): foundation design, construction sequencing/scheduling carbon, operational (in-use) carbon — embodied carbon is scoped to material production + construction (cradle-to-site) only.

6. Technical Workflow

Five phases, each with a concrete exit criterion — the project does not proceed to the next phase until the current one's exit criterion is met.

Phase 0 — Problem Formulation

Finalize the variable list, discrete/continuous ranges, and every hard constraint (deflection limit, utilization ≤ 1, minimum/maximum reinforcement ratio per IS 456, bar spacing, and ductile detailing rules if seismic is in scope). This is written down before any code is run.

Exit criterion: signed-off variable and constraint table (Section 8), reviewed by guide.

Phase 1 — Design of Experiments (Sampling)

Generate the initial training sample using Latin Hypercube Sampling (LHS) over the continuous variables, combined with sampling from actual discrete option lists for section sizes and bar diameters (not continuous approximations of discrete IS-standard sizes). Target: ~2,000–5,000 initial design combinations.

Exit criterion: sampled design matrix generated and spot-checked for physically sensible combinations (no negative effective depth, no reinforcement ratio outside IS 456 bounds, etc.).

Phase 2 — Finite-Element Data Generation

Batch-run OpenSeesPy on the sampled design matrix, recording deflection, member utilization ratio, and failure/non-convergence flags for each. This is the expensive, one-time data-generation cost the rest of the pipeline is built to avoid repeating.

Exit criterion: complete labeled dataset (inputs → structural response) with failed/non-converged runs logged and understood, not silently dropped.

Phase 3 — Surrogate Model Training

Train a regression model (e.g. XGBoost or gradient-boosted trees) per constraint — deflection and utilization ratio — not a single pass/fail classifier, so the optimizer has a continuous constraint-violation signal rather than a hard cliff. Validate on a held-out split and report error metrics honestly, including where the surrogate is weakest (typically near the feasibility boundary).

Exit criterion: surrogate validation error reported and judged acceptable by guide before it is trusted inside the optimization loop.

Phase 4 — Constrained Multi-Objective Optimization

Run constrained NSGA-II (via pymoo) with cost and embodied carbon as the two objectives, using the Phase 3 surrogates to evaluate structural constraints at each candidate design — this is what makes thousands of NSGA-II evaluations computationally feasible. Output: a Pareto front of non-dominated designs.

Exit criterion: converged, non-dominated Pareto front produced and visualized.

Phase 5 — Verification & Reporting

Re-run real OpenSeesPy analysis on a representative subset (proposed: 20–30 designs) spanning the Pareto front to confirm the surrogate did not select infeasible designs. Report any surrogate/ground-truth discrepancies found, and the final validated Pareto front becomes the project's headline result.

Exit criterion: verification results reported — including any designs that fail ground-truth re-check — not omitted for a cleaner narrative.

7. Tools & Technology Stack

Component	Tool	Rationale
Structural FE analysis	OpenSeesPy	Scriptable, headless, fast repeated analysis — suited to batch data generation. STAAD.Pro deliberately excluded from the automated loop (no practical Python-native API for thousands of iterative calls).
Sampling / DOE	Python (scipy / pyDOE)	Latin Hypercube Sampling over the design space.
Surrogate model	XGBoost / scikit-learn	Fast, well-understood regression on tabular structural response data.
Optimization	pymoo (NSGA-II, constrained)	Native constrained-domination handling suited to the deflection/utilization constraints.
Embodied carbon data	ICE database (baseline) + Indian EPDs / supplier data where available	ICE used as a documented baseline; Indian-context factors sourced separately and gaps explicitly flagged rather than silently assumed.
Verification	OpenSeesPy (targeted re-runs)	Ground-truth check on final Pareto-selected designs.

8. Design Variables & Constraints (Draft — for Guide's Sign-off)

Design variables

Variable	Type	Range / options (proposed)
Beam width, depth	Discrete (IS standard sizes)	230–450 mm width, 300–750 mm depth, in standard increments
Column dimensions	Discrete (IS standard sizes)	230–600 mm, square/rectangular options
Longitudinal reinforcement ratio	Continuous, bounded	Per IS 456 min/max limits for beams and columns
Rebar diameter	Discrete	IS standard bar sizes (10, 12, 16, 20, 25 mm etc.)
Concrete grade	Discrete	M20, M25, M30, M35, M40
Steel grade	Discrete	Fe415, Fe500

Hard constraints

- Max deflection ≤ span/250 (or applicable IS 456 limit) — serviceability.
- Member utilization ratio ≤ 1.0 under IS 456 / IS 800 limit state design — strength.
- Reinforcement ratio within IS 456 minimum/maximum bounds.
- Bar spacing and detailing within code limits (ductile detailing per IS 13920 if seismic scope is confirmed).
- All discrete choices (section sizes, bar diameters, grades) restricted to real, commercially available IS-standard options — no continuous relaxation presented as buildable.

9. Deliverables

#	Deliverable	Phase
1	Signed-off variable/constraint specification	Phase 0
2	Labeled FE dataset (design inputs → structural response)	Phase 2
3	Trained + validated surrogate model with reported error metrics	Phase 3
4	Pareto front (cost vs. embodied carbon) with visualization	Phase 4
5	Verification report (surrogate vs. ground-truth OpenSeesPy)	Phase 5
6	Final project report + codebase (reproducible pipeline)	Final

10. Timeline & Milestones

Proposed semester-wise breakdown — adjustable against the guide's own major-project calendar and evaluation checkpoints.

Milestone	Target window	Output
Phase 0 — Formulation & literature review	Weeks 1–3	Signed-off scope document (this brief, finalized)
Phase 1 — Sampling pipeline built	Weeks 4–5	LHS + discrete sampler, test batch generated
Phase 2 — Full FE data generation run	Weeks 6–9	Complete labeled dataset
Mid-evaluation checkpoint	~Week 9–10	Dataset + Phase 0–2 results presented
Phase 3 — Surrogate training & validation	Weeks 10–12	Validated regression surrogates
Phase 4 — NSGA-II optimization runs	Weeks 13–15	Pareto front(s) generated
Phase 5 — Verification & report writing	Weeks 16–18	Verified Pareto front, draft report
Final submission & viva prep	Weeks 19–20	Final report, codebase, presentation

11. Risk Register

Risk	Likelihood	Mitigation
Indian-context embodied carbon data unavailable or inconsistent	High	Use ICE as documented baseline; explicitly flag which factors are India-adapted vs. assumed; treat this gap itself as a reportable finding.
OpenSeesPy non-convergence on edge-case sampled designs	Medium	Log and exclude non-converged runs with reasons documented, don't silently drop or interpolate over them.
Surrogate accuracy poor near the feasibility boundary (exactly where Pareto-optimal designs cluster)	Medium–High	Concentrate verification re-runs (Phase 5) on boundary designs specifically; consider one round of active-learning resampling if time allows.
Scope creep (seismic detailing, multiple building typologies)	Medium	Seismic and multi-typology explicitly deferred to Section 5 out-of-scope list unless guide requires otherwise.
Timeline compression against concurrent academic backlog	Medium	Phase exit criteria (Section 6) allow early detection of slippage rather than discovering it at final submission.

12. Resources & Support Requested from Faculty Guide

- Confirmation/adjustment of the scope defaults in Section 5 (typology, seismic in/out, building height range).
- Guidance or access to any departmental reference data on Indian material embodied-carbon factors, if available, in place of defaulting to ICE.
- Sign-off on the design variable and constraint table (Section 8) before Phase 1 begins.
- Scheduled checkpoint reviews aligned to the Section 10 timeline, particularly the Phase 2→3 and Phase 4→5 handoffs where validation matters most.
- Any institutional compute access (if departmental machines are available) — the batch OpenSeesPy runs in Phase 2 are the most compute-intensive step.

13. Success / Evaluation Criteria

- A non-dominated Pareto front is produced and at least 90% of the re-verified subset (Phase 5) matches surrogate predictions within an agreed tolerance.
- Surrogate model achieves validation error the guide judges acceptable for design-stage (not final-design) use.
- Final report honestly documents surrogate/ground-truth discrepancies rather than presenting only favorable results.
- Codebase is reproducible — another student could re-run the pipeline end-to-end from the report and code alone.

14. Reference Points (Indicative)

Cited as prior art establishing the technique's maturity, not as sources the project claims to extend algorithmically:

- Kaveh, A. et al., “Multi-objective optimization of reinforced concrete frames using NSGA-II algorithm,” Journal of Civil Engineering and Management, 2016.
- ML-surrogate NSGA-II framework for cost/strength/carbon optimization in nano-silica-modified concrete mix design (recent, indicative of the surrogate-assisted pattern applied to a related materials problem).
- IS 456:2000 — Plain and Reinforced Concrete — Code of Practice.
- IS 800:2007 — General Construction in Steel — Code of Practice.
- IS 875 (Parts 1–5) — Design Loads for Buildings and Structures.
- ICE Database (Inventory of Carbon and Energy), University of Bath — baseline embodied carbon factors.

End of brief — open items for guide's decision are marked in Sections 5, 8 and 12.